

Mohamed Yassin

DevOps Engineer

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About Me

Fourth-year Computer Science student with hands-on experience in Linux systems, backend development, and cloud-native DevOps. Skilled with Docker, Kubernetes, GCP, Python, CI/CD (GitHub Actions, Jenkins) and container networking — built and operated production services, automated deployments, and troubleshoot distributed systems. Experienced in designing scalable infrastructure, production support, observability and data-quality pipelines. Active competitive programmer with interests in cybersecurity and system design.

Work Experience

Intern – *Egyptian Space Agency (EgSA), Earth Observation Data Department* Aug 2025 – Nov 2025

- Built backend services for the Egyptian Heritage Satellite Atlas using TypeScript and Google Earth Engine to process satellite imagery for urban analysis and land-use mapping.
- Managed Google Cloud Platform infrastructure on Linux, handling deployments, maintenance, and production troubleshooting to maintain service availability.
- Implemented CI/CD pipelines with GitHub Actions to automate builds and deployments, reducing manual release steps and accelerating updates.

Projects

PeerLink – Distributed File Sharing Platform [repo](#) [🔗](#)

- Built a distributed, fault-tolerant file-sharing platform using **TypeScript, Node.js, React, Docker, Redis, and PostgreSQL**.
- Developed a microservices backend with **Redis Cluster** coordination and **Prisma/PostgreSQL** persistence, implementing leader election and high-availability mechanisms.

HoneyKube – AI-Powered Kubernetes Honeypot [repo](#) [🔗](#)

- Developed a scalable LLM-powered honeypot using **Python, Kubernetes, Docker, Redis, and OpenRouter** to emulate vulnerable services and collect telemetry.
- Designed and maintained cluster infrastructure, troubleshooting container networking and orchestration issues to ensure stable operation.

GUIShell – High-Performance Remote Desktop Engine [repo](#) [🔗](#)

- Built a low-latency remote desktop platform using **Electron, WebRTC, and Node.js**, featuring a custom MJPEG-over-DataChannel streaming protocol.
- Developed a signaling server with **SQLite** persistence and automated packaging and release pipelines using **Electron Builder**.

CCNA 200-301 Final Project [repo](#) [🔗](#)

- Designed and deployed an enterprise network using **Cisco Packet Tracer**, implementing VLANs, DNS, EtherChannel, STP, subnetting, and OSPF.

Key Achievements

- **TurboNodeIO** [🔗](#) – Built a native C++ acceleration layer for Node.js, achieving up to **317× faster** I/O performance and **71× faster** SIMD checksum processing.
- **1st Place, Google Hackathon (College)** – Developed an embedded assistive device that helps visually impaired users pour water safely using sensor-based detection.

Technical Skills

Programming: C++, TypeScript, JavaScript, Python, Bash

Backend & Databases: Node.js, Microservices, Distributed Systems, REST APIs, PostgreSQL, SQL, Redis, Prisma ORM

Frontend: React, Electron

DevOps & Cloud: Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), GCP (Google Earth Engine), Linux Administration, Nginx, Git, SSH, Container Networking

Networking & Security: OSPF, VLANs, STP, EtherChannel, Subnetting, Penetration Testing, Web Security, Network Security

Education

Modern Academy – *B.S. in Computer Science*

Sept 2022 – Present

Additional Skills

Languages: Arabic (Native), English (Fluent)

Certifications: CCNA 200-301

Activities: Competitive Programming (Codeforces, LeetCode), Problem Solving, ECPC Participant, ICMTCTF Participant

Soft Skills: Leadership, Problem-Solving, Teamwork, Self-Learning, Adaptability